Aim : Apply COCOMO Model for cost estimation

Objective : Understanding the COCOMO model through one of the example.

Code :

```
import java.util.Arrays;

public class BasicCOCOMO {
    private static final double[][] TABLE = {
        {2.4, 1.05, 2.5, 0.38},
        {3.0, 1.12, 2.5, 0.35},
        {3.6, 1.20, 2.5, 0.32}
    };
    private static final String[] MODE = {
        "Organic", "Semi-Detached", "Embedded"
    };

    public static void calculate(int size) {
        int model = 0;

        // Check the mode according to size
        if (size >= 2 && size <= 50) {
            model = 0;
        } else if (size > 50 && size <= 300) {
            model = 1;
        } else if (size > 300) {
            model = 2;
        }

        System.out.println("The mode is " + MODE[model]);

        // Calculate Effort
        double effort = TABLE[model][0] * Math.pow(size, TABLE[model][1]);
```

```
        // Calculate Time
        double time = TABLE[model][2] * Math.pow(effort, TABLE[model][3]);

        // Calculate Persons Required
        double staff = effort / time;

        // Output the values calculated
        System.out.println("Effort = " + Math.round(effort) + " Person-Month");
        System.out.println("Development Time = " + Math.round(time) + " Months");
        System.out.println("Average Staff Required = " + Math.round(staff) + " Persons");
    }
    public static void main(String[] args) {
        int size = 4;
        calculate(size);
    }
}
```

Output:

The mode is Organic

Effort = 10 Person-Month

Development Time = 6 Months

Average Staff Required = 2 Persons

Conclusion : Using above code we are abled to calculate the Effort & Development time of the project.